DATA ANALYSIS INTERNSHIP REPORT

Cognifyz Technologies

Student Details

Name: Partho Maji

Domain: Data Analysis Internship

Tools Used: Python, Pandas, Matplotlib

LEVEL 1 TASKS

- Task 1: Top Cuisines

Objective

To identify the most common cuisines and their distribution.

Analysis

Extracted cuisines and separated multiple values using splitting and exploding techniques.

Identified the top 3 most common cuisines.

Calculated percentage contribution of each cuisine.

Conclusion

The dataset shows that a few cuisines dominate the market, indicating strong customer preference for specific food types.

- Task 2: City Analysis

Objective

To analyze restaurant distribution and ratings across cities.

Analysis

Identified the city with the highest number of restaurants.

Calculated average ratings per city.

Determined the highest-rated city.

Conclusion

Major cities have a higher number of restaurants, while some cities maintain better average ratings, indicating quality differences across regions.

- 🔹 Task 3: Price Range Distribution

Objective

To understand how restaurants are distributed across different price ranges.

Analysis

Categorized restaurants into 4 price levels:

1 → Low Cost

2 → Moderate

3 → Expensive

4 → Luxury

Visualized distribution using bar and pie charts.

Conclusion

Most restaurants fall into the moderate price category, suggesting that customers prefer affordable dining options.

- 🔹 Task 4: Online Delivery

Objective

To analyze the impact of online delivery on restaurant ratings.

Analysis

Calculated percentage of restaurants offering online delivery.

Compared average ratings of restaurants with and without delivery.

Conclusion

Restaurants offering online delivery tend to have slightly higher ratings, indicating that convenience plays a significant role in customer satisfaction.

LEVEL 2 TASKS

- Task 1: Restaurant Ratings

Objective

To analyze rating distribution and customer engagement.

Analysis

Removed unrated entries (rating = 0).

Plotted histogram of ratings.

Categorized ratings into:

Poor, Average, Good, Excellent

Calculated average votes.

Conclusion

Most restaurants fall in the “Good” to “Excellent” category, indicating overall positive service quality. The average number of votes shows moderate customer engagement.

- Task 2: Cuisine Combination

Objective

To analyze popular cuisine combinations and their ratings.

Analysis

Identified most frequent cuisine combinations.

Calculated average ratings for each combination.

Applied filtering (minimum count threshold) to avoid biased results.

Conclusion

Popular cuisine combinations such as multi-cuisine offerings are common. While some rare combinations have very high ratings, reliable insights come from frequently occurring combinations.

- Task 3: Geographic Analysis

Objective

To analyze spatial distribution of restaurants.

Analysis

Plotted restaurant locations using latitude and longitude.

Observed clustering patterns in specific areas.

Conclusion

Restaurants are highly concentrated in urban and commercial areas, forming clear clusters. Sparse distribution in outskirts indicates lower demand or population density. Clustering reflects high competition and customer demand zones.

- Task 4: Restaurant Chains

Objective

To analyze performance of restaurant chains.

Analysis

Identified chains based on repeated restaurant names.

Measured popularity using number of outlets.

Calculated average ratings for chains.

Filtered chains with sufficient outlets for reliable analysis.

Conclusion

Some chains show strong brand presence with many outlets, while others maintain high ratings. The best-performing chains balance expansion with consistent service quality.

FINAL CONCLUSION

This analysis highlights key insights into restaurant trends:

Customer preferences are concentrated around specific cuisines and moderate pricing.

Urban areas dominate restaurant distribution, forming competitive clusters.

Online delivery enhances customer satisfaction and ratings.

Successful restaurant chains maintain both scalability and consistent quality.

Overall, the dataset reflects a competitive and customer-driven food industry where location, pricing, and service quality are key success factors.